



Software Testing

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What is Testing



- ✧ Testing is a group of techniques to determine the correctness of the application under the predefined script but, testing cannot find all the defect of application.
- ✧ The main intent of testing is to detect failures of the application so that failures can be discovered and corrected.
- ✧ It does not demonstrate that a product functions properly under all conditions but only that it is not working in some specific conditions.

Software Testing



- ✧ Software testing is widely used technology because it is compulsory to test each and every software before deployment.
- ✧ Software testing is a process of identifying the correctness of software by considering its all attributes (Reliability, Scalability, Portability, Re-usability, Usability) and evaluating the execution of software components to find the software bugs or errors or defects.

Attributes of Software



Reliability

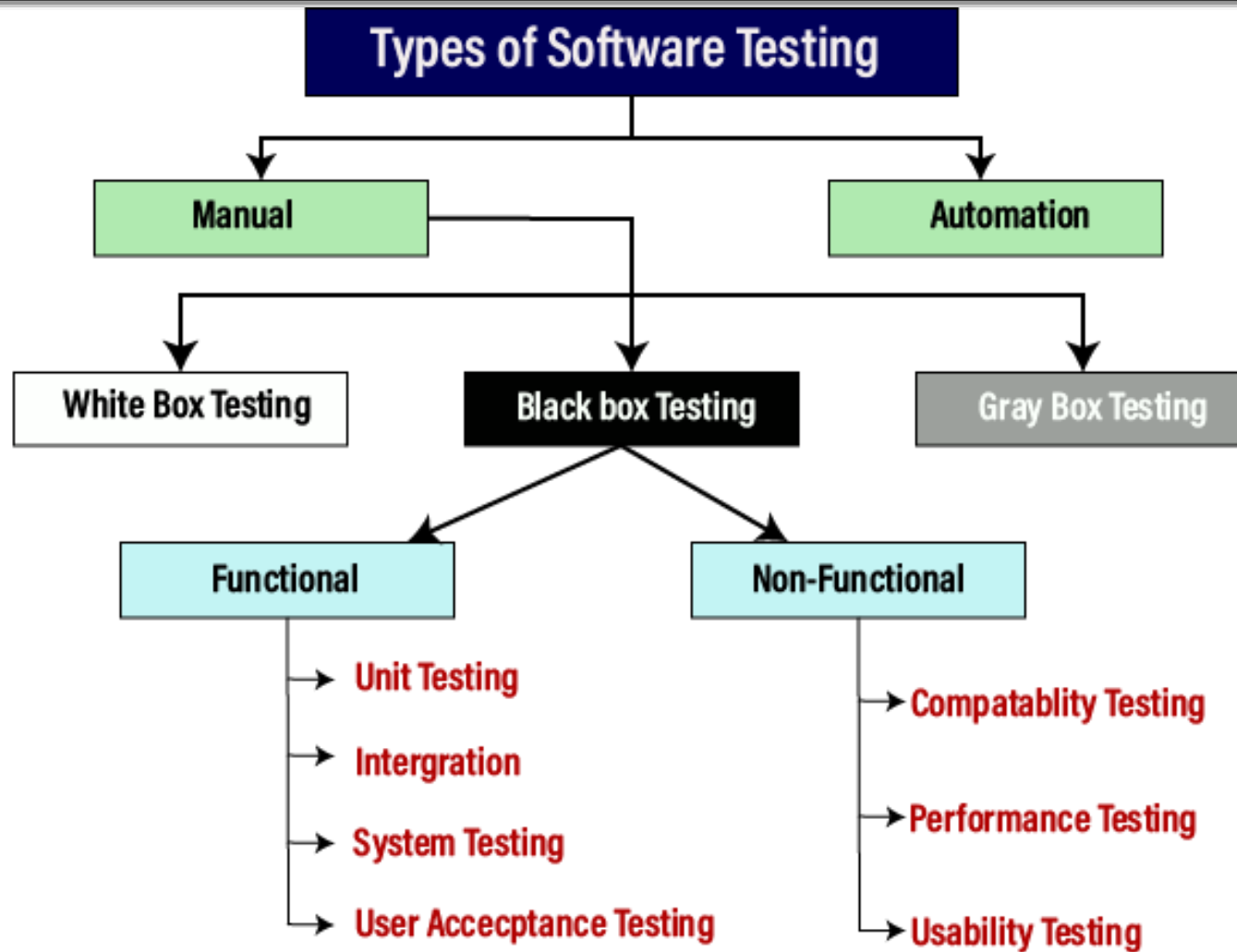
Scalability

Portability

Reusability

Usability

Type of Software testing



Automation testing



- ✧ Automation testing is a process of converting any manual test cases into the test scripts with the help of automation tools, or any programming language is known as automation testing.
- ✧ With the help of automation testing, we can enhance the speed of our test execution because here, we do not require any human efforts.
- ✧ We need to write a test script and execute those scripts.

WHAT IS AUTOMATION TESTING ?

TOOLS

FRAMEWORKS



Script



Software



Result

Manual testing



- ✧ The process of checking the functionality of an application as per the customer needs without taking any help of automation tools is known as manual testing.
- ✧ While performing the manual testing on any application, we do not need any specific knowledge of any testing tool, rather than have a proper understanding of the product so we can easily prepare the test document.
- ✧ Manual testing can be further divided into three types of testing
 - **White box testing**
 - **Black box testing**
 - **Gray box testing**

White Box Testing



- ✧ The box testing approach of software testing consists of black box testing and white box testing.
- ✧ We are discussing here white box testing which also known as glass box is **testing, structural testing, clear box testing, open box testing and transparent box testing**
- ✧ The term 'white box' is used because of the internal perspective of the system.
- ✧ The clear box or white box or transparent box name denote the ability to see through the software's outer shell into its inner workings.

Types of White Box Testing



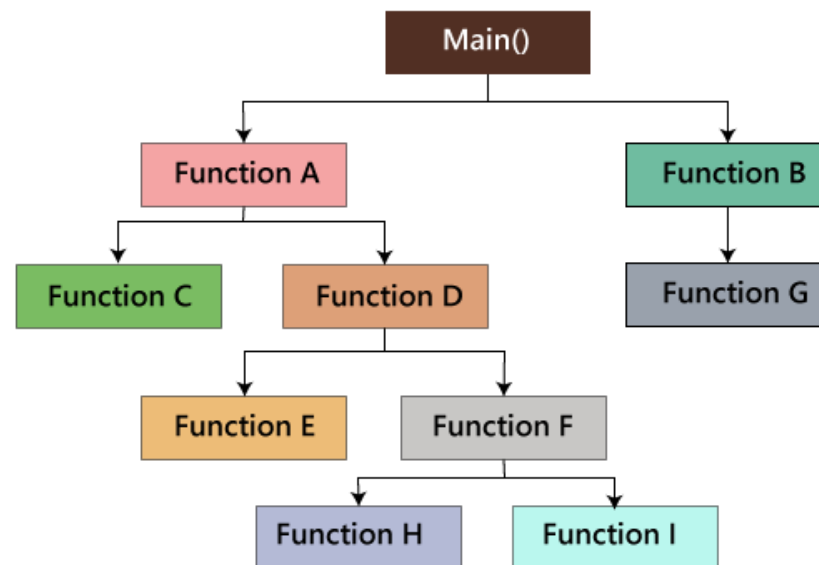
✧ The white box testing contains various tests, which are as follows

- Path testing
- Loop testing
- Condition testing
- Testing based on the memory perspective
- Test performance of the program

Path testing



- ✧ In the path testing, we will write the flow graphs and test all independent paths.
- ✧ Here writing the flow graph implies that flow graphs are representing the flow of the program and also show how every program is added with one another



Loop testing



- ✧ In the loop testing, we will test the loops such as while, for, and do-while, etc. and also check for ending condition if working correctly and if the size of the conditions is enough.
- ✧ **For example:** we have one program where the developers have given about 50,000 loops.

```
{  
while(50,000)  
.....  
.....  
}
```

Loop Testing Example



- ✧ We cannot test this program manually for all the 50,000 loops cycle.
- ✧ So we write a small program that helps for all 50,000 cycles, as we can see in the below program, that test P is written in the similar language as the source code program, and this is known as a Unit test.
- ✧ And it is written by the developers only.

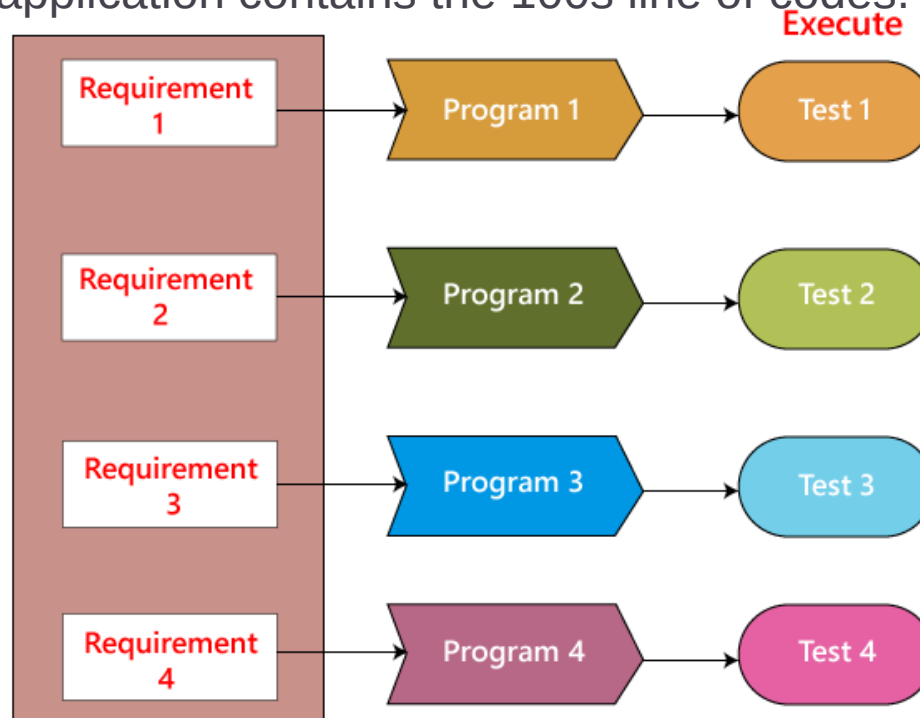
Test P

```
{  
.....  
..... }
```

Loop Testing Example



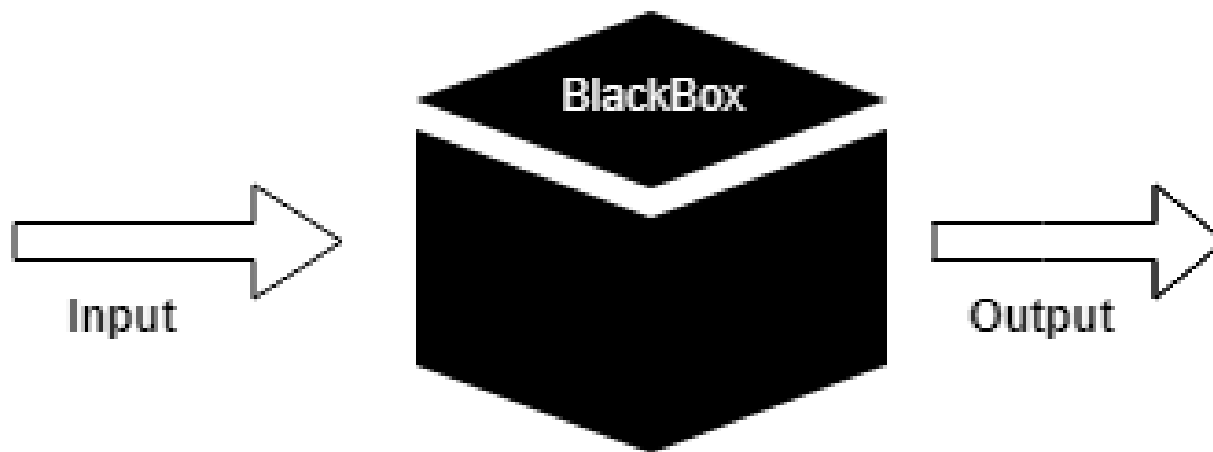
- ✧ As we can see in the below image that, we have various requirements such as 1, 2, 3, 4. And then, the developer writes the programs such as program 1,2,3,4 for the parallel conditions.
- ✧ Here the application contains the 100s line of codes.



Black box testing



- ✧ Black box testing is a technique of software testing which examines the functionality of software without peering into its internal structure or coding.
- ✧ The primary source of black box testing is a specification of requirements that is stated by the customer.



Generic steps of black box testing



- ✧ The black box test is based on the specification of requirements, so it is examined in the beginning.
- ✧ In the second step, the tester creates a positive test scenario and an adverse test scenario by selecting valid and invalid input values to check that the software is processing them correctly or incorrectly.
- ✧ In the third step, the tester develops various test cases such as decision table, all pairs test, equivalent division, error estimation, cause-effect graph, etc.
- ✧ The fourth phase includes the execution of all test cases.
- ✧ In the fifth step, the tester compares the expected output against the actual output.
- ✧ In the sixth and final step, if there is any flaw in the software, then it is cured and tested again.

Functional Testing



- ✧ It is a type of software testing which is used to verify the functionality of the software application, whether the function is working according to the requirement specification.
- ✧ In functional testing, each function tested by giving the value, determining the output, and verifying the actual output with the expected value.
- ✧ Functional testing performed as black-box testing which is presented to confirm that the functionality of an application or system behaves as we are expecting.

Goal of functional testing



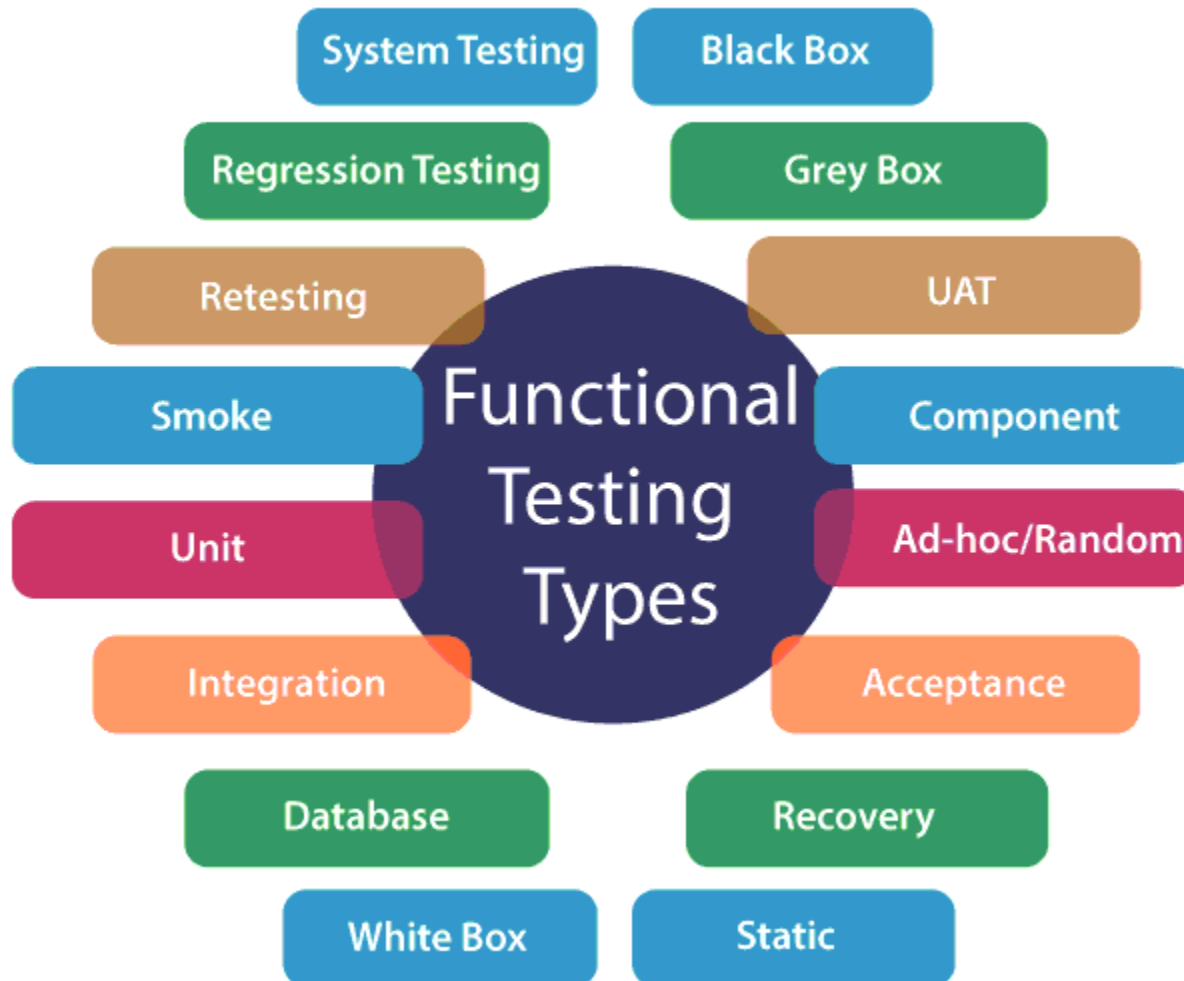
- ✧ The purpose of the functional testing is to check the primary entry function, necessarily usable function
- ✧ Testers follow the following steps in the functional testing:
 - Tester does verification of the requirement specification in the software application.
 - After analysis, the requirement specification tester will make a plan.
 - After planning the tests, the tester will design the test case.
 - After designing the test, case tester will make a document of the traceability matrix.
 - The tester will execute the test case design.
 - Analysis of the coverage to examine the covered testing area of the application.
 - Defect management should do to manage defect resolving.

What to test in functional testing?



- ✧ The main objective of functional testing is checking the functionality of the software system. It concentrates on:
 - **Basic Usability:** Functional Testing involves the usability testing of the system. It checks whether a user can navigate freely without any difficulty through screens.
 - **Accessibility:** Functional testing test the accessibility of the function.
 - **Mainline function:** It focuses on testing the main feature.
 - **Error Condition:** Functional testing is used to check the error condition. It checks whether the error message displayed.

Types of Functional Testing



Unit Testing (Functional Testing)



- ✧ **Unit testing** is a type of software testing, where the individual unit or component of the software tested.
- ✧ Unit testing, examine the different part of the application, by unit testing functional testing also done, because unit testing ensures each module is working correctly.
- ✧ The developer does unit testing.
- ✧ Unit testing is done in the development phase of the application.

Smoke Testing (Functional Testing)



- ✧ **Functional testing** by smoke testing. Smoke testing includes only the basic (feature) functionality of the system.
- ✧ Smoke testing is known as "***Build Verification Testing.***"
- ✧ Smoke testing aims to ensure that the most important function work.
- ✧ For example, Smoke testing verifies that the application launches successfully will check that GUI is responsive.

Sanity Testing (Functional Testing)



- ✧ **Sanity testing** involves the entire high-level business scenario is working correctly.
- ✧ Sanity testing is done to check the functionality/bugs fixed. Sanity testing is little advance than smoke testing.
- ✧ For example,
 - login is working fine;
 - all the buttons are working correctly;
 - after clicking on the button navigation of the page is done or not.

Database Testing (Functional Testing)



- ✧ Database testing is a type of testing which checks the schema, tables, triggers, etc. of the database under test.
- ✧ Database testing may involve creating complex queries to load/stress test the database and check its responsiveness.
- ✧ It checks the data integrity and consistency.

Non-Functional Testing



- ✧ Non-functional testing is a type of software testing to test non-functional parameters such as reliability, load test, performance and accountability of the software.
- ✧ The primary purpose of non-functional testing is to test the reading speed of the software system as per non-functional parameters.
- ✧ The parameters of non-functional testing are never tested before the functional testing
- ✧ For example, non-functional testing would be to test how many people can work simultaneously on any software.

Types of Non Functional Testing



Load Testing (Non Functional Testing)



- ✧ Load testing involves testing the system's loading capacity.
- ✧ Loading capacity means more and more people can work on the system simultaneously.

Security Testing (Non Functional Testing)



- ✧ Security testing is used to detect the security flaws of the software application.
- ✧ The testing is done via investigating system architecture and the mindset of an attacker.
- ✧ Test cases are conducted by finding areas of code where an attack is most likely to happen.

Portability Testing (Non Functional Testing)



- ✧ The portability testing of the software is used to verify whether the system can run on different operating systems without occurring any bug.
- ✧ This test also tests the working of software when there is a same operating system but different hardware.

Accountability Testing (Non Functional Testing)



- ✧ Accountability test is done to check whether the system is operating correctly or not.
- ✧ A function should give the same result for which it has been created.
- ✧ If the system gives expected output, it gets passed in the test otherwise failed.

Reliability Testing (Non Functional Testing)



- ✧ Reliability test assumes that whether the software system is running without fail under specified conditions or not.
- ✧ The system must be run for a specific time and number of processes. If the system is failed under these specified conditions, reliability test will be failed.

Efficiency Testing (Non Functional Testing)

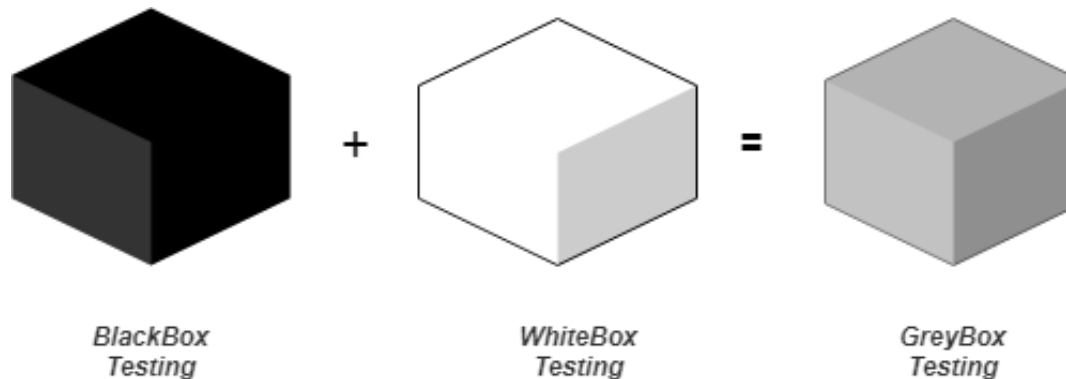


- ✧ Efficiency test examines the number of resources needed to develop a software system, and how many of these were used.
- ✧ It also includes the test of these three points.
 - Customer's requirements must be satisfied by the software system.
 - A software system should achieve customer specifications.
 - Enough efforts should be made to develop a software system.

GreyBox Testing



- ✧ Greybox testing is a software testing method to test the software application with partial knowledge of the internal working structure.
- ✧ It is a **combination of black box and white box testing** because it involves access to internal coding to design test cases as white box testing and testing practices are done at functionality level as black box testing.



Why GreyBox testing?



- ✧ It provides combined benefits of both Blackbox testing and WhiteBox testing.
- ✧ It includes the input values of both developers and testers at the same time to improve the overall quality of the product.
- ✧ It reduces time consumption of long process of functional and non-functional testing.
- ✧ It gives sufficient time to the developer to fix the product defects.
- ✧ It includes user point of view rather than designer or tester point of view.
- ✧ It involves examination of requirements and determination of specifications by user point of view deeply.

GreyBox Testing Strategy



- ✧ Grey box testing does not make necessary that the tester must design test cases from source code.
- ✧ To perform this testing test cases can be designed on the base of, knowledge of architectures, algorithm, internal states or other high -level descriptions of the program behavior.
- ✧ It uses all the straightforward techniques of black box testing for function testing.

GreyBox Testing Strategy



✧ Generic Steps to perform Grey box Testing are:

- First, select and identify inputs from BlackBox and WhiteBox testing inputs.
- Second, Identify expected outputs from these selected inputs.
- Third, identify all the major paths to traverse through during the testing period.
- The fourth task is to identify sub-functions which are the part of main functions to perform deep level testing.
- The fifth task is to identify inputs for subfunctions.
- The sixth task is to identify expected outputs for subfunctions.
- The seventh task includes executing a test case for Subfunctions.
- The eighth task includes verification of the correctness of result.

Techniques of Grey box Testing



✧ Matrix Testing

- This testing technique comes under Grey Box testing.
- It defines all the used variables of a particular program.
- In any program, variable are the elements through which values can travel inside the program.
- It should be as per requirement otherwise, it will reduce the readability of the program and speed of the software.
- Matrix technique is a method to remove unused and uninitialized variables by identifying used variables from the program.

Techniques of Grey box Testing



✧ Regression Testing

- Regression testing is used to verify that modification in any part of software has not caused any adverse or unintended side effect in any other part of the software.
- During confirmation testing, any defect got fixed, and that part of software started working as intended, but there might be a possibility that fixed defect may have introduced a different defect somewhere else in the software.
- So, regression testing takes care of these type of defects by testing strategies like retest risky use cases, retest within a firewall, retest all, etc.